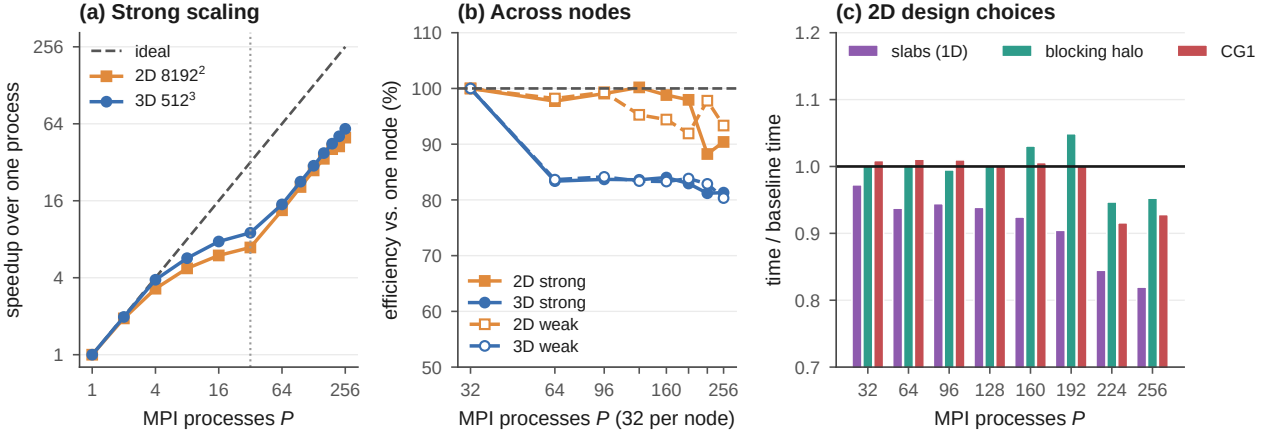
(a) Strong scaling
ideal
2D 8192²
3D 512³
speedup over one process
256
64
16
4
1
1
4
16
64
256
MPI processes P
(b) Across nodes
efficiency vs. one node (%)
110
100
90
80
70
60
50
2D strong
3D strong
2D weak
3D weak
32
64
96
160
256
MPI processes P (32 per node)
(c) 2D design choices
slabs (1D)
blocking halo
CG1
time / baseline time
1.2
1.1
1.0
0.9
0.8
0.7
32
64
96
128
160
192
224
256
MPI processes P